



# Scaling in Underworld

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## Abstract

To test scalability we run weak scaling tests on various HPC machines to check the numerical framework remains robust when pushing for higher fidelity models.

**Keywords** Documentation, Underworld Code

To date *weak scaling* tests have been run on two of the largest computers in Australia: **Gadi** (NCI) and **Magnus** (Pawsey).

Here we present the results of those tests and discuss:

## 1. GADI: WEAK SCALING - SOLDB3D Q1

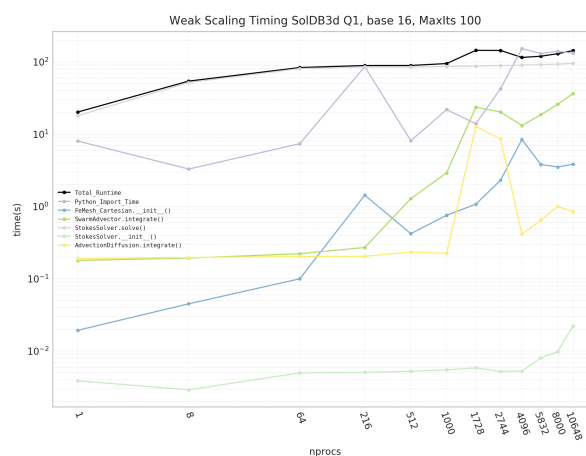


Figure 1: Gadi - v2.11-prerelease: Weak scaling of SolDB3D (linear elements) to ~10k procs

## 2. GADI: WEAK SCALING - SOLDB3D Q2

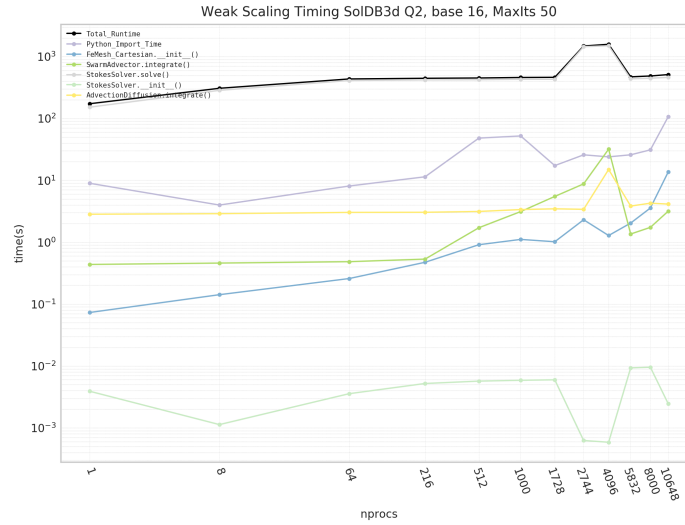


Figure 2: Gadi - v2.11-prerelease: Weak scaling of SolDB3D (quadratic elements) to 10k procs

### 3. MAGNUS: WEAK SCALING - SOLDB3D Q2 - v2.10 vs v2.9

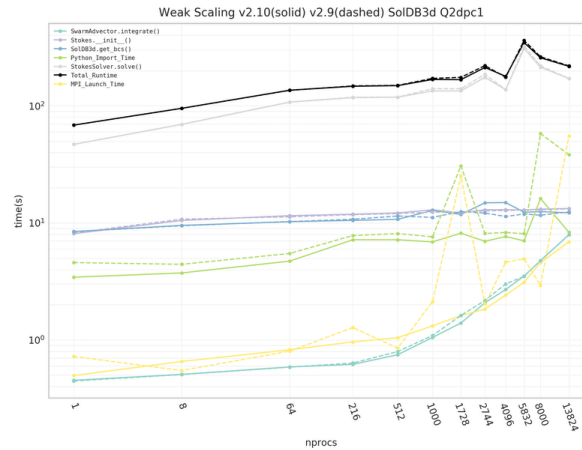


Figure 3: Magnus: Weak scaling of SolDB3D (quadratic elements) to ~14k procs

Underworld's **Gadi** installation is setup as a "bare metal" install, i.e. all code and dependencies are natively compiled onto Gadi's filesystem. However **Magnus** utilises Underworld's custom prebuilt docker via singularity containerisation.

Generally we see Underworld can scale to beyond 10k CPUs on both **Gadi** and **Magnus**. Wonderful!

It is noticeable that the *Python\_Import\_Time* scales more erratically on all Gadi runs rather than Mangus runs. Indeed it has been observed that some Gadi jobs, 500+ CPUs, fail to even start.

It is believed this issue is related to Underworld's (many) Python modules being read concurrently from Gadi's filesystem, overloading the filesystem with metadata operations and blocking IO. While on Magnus all Python modules are available to every CPU via the docker container and *Python\_Import\_Time* scales consistently.

The Underworld development team are continuing to work with the Gadi system admin to overcome this *Python\_Import\_Time* issue.

The analytic solution **SolDB3D** used as the reference model is based on the work by Dohrmann, C & Bochev, Pavel. 2004 [1], for the implementation details of SolDB3D in Underworld see [here](#)

[1]: Dohrmann, C & Bochev, Pavel. (2004). A stabilized finite element method for the Stokes problem based on polynomial pressure projections. *International Journal for Numerical Methods in Fluids*. 46. 183-201.